

Baseline Model (Model 0)

Assignment: For this model, use displacement-based elements. Adjust the length of elements next to the beam-column joint, referred to as the plastic hinge elements, to be one half of the depth of the associated section. For other elements, use a comparable length. Use 5 Gauss-Lobatto integration points per element. You should get a reasonable agreement between the numerical and experimental results similar to Figure 3.

Codes:

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/Model_0.py

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/Model_0_plotting.py

Observations:

Comparing the Model 0 numerical response with the experimental curve, the results are similar to that of the expected curve (Figure 3). The model reasonably captures the overall lateral response, particularly the peak lateral strength of the beam-column subassembly. However, there is a significant discrepancy in the shape and width of the cyclic loops. The experimental curve shows the Pinching effect, as there is narrowing of the hysteresis loop curve near the center during the cyclic loading.

In contrast, Model 0 produces much wider hysteresis loops because the Steel02 and Concrete02 material models do not account for stiffness degradation or pinching behavior. Thus, it overestimates the energy the joint can actually dissipate.

Overall, the real test specimen showed more stiffness degradation and softened much faster during repeated cycles than the baseline numerical model.

Conclusion:

Model 0 captures the overall lateral strength and general hysteresis envelope reasonably well, confirming the basic force-deformation capacity of the subassembly. However, it significantly overestimates energy dissipation, producing much wider loops than observed experimentally. The experimental specimen showed greater stiffness degradation and pinching, which is not reproduced by the baseline model.

Displacement-based elements: 9 (left/right) and 5 (top/bottom); ≈ 200 mm each

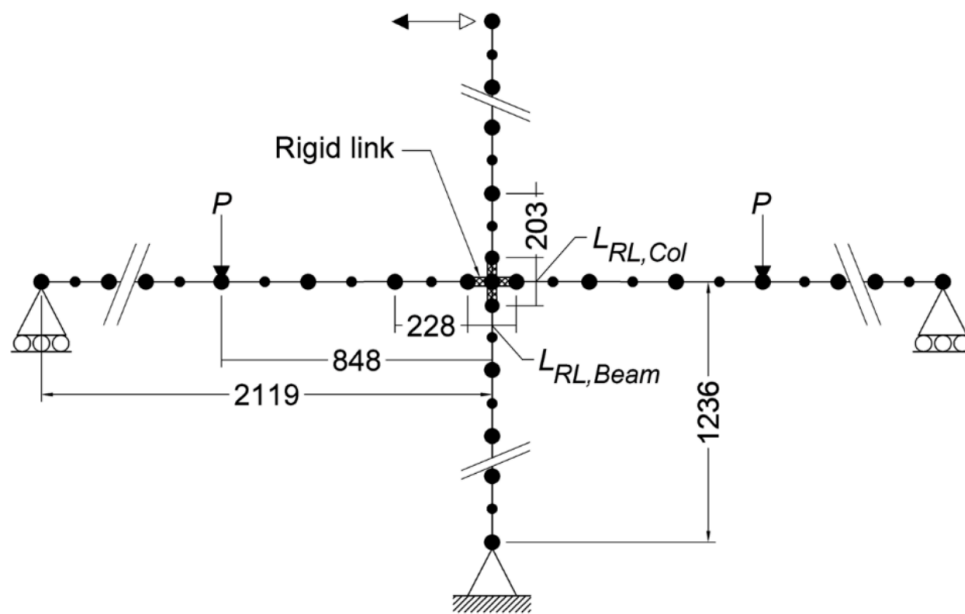


Figure 1. FE modeling

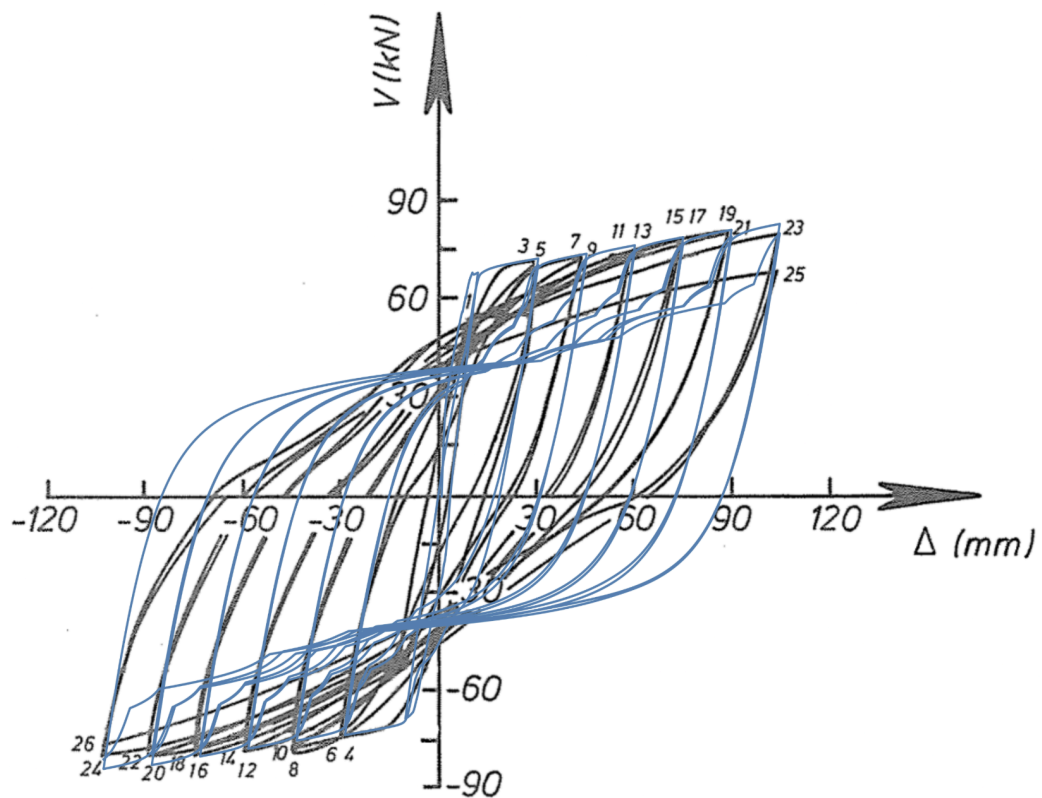


Figure 2. Observed numerical (blue) - Experimental (black) correlation of Model 0

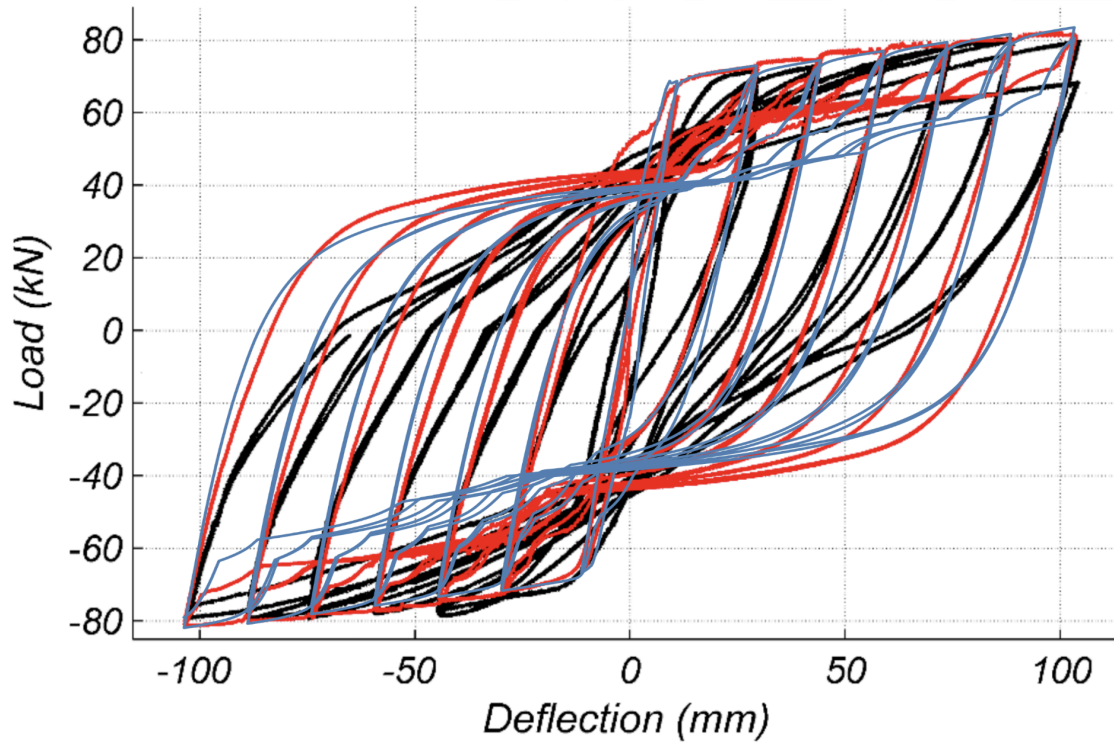


Figure 3. Expected numerical (red), Observed numerical (blue), Experimental (black) correlation of Model 0

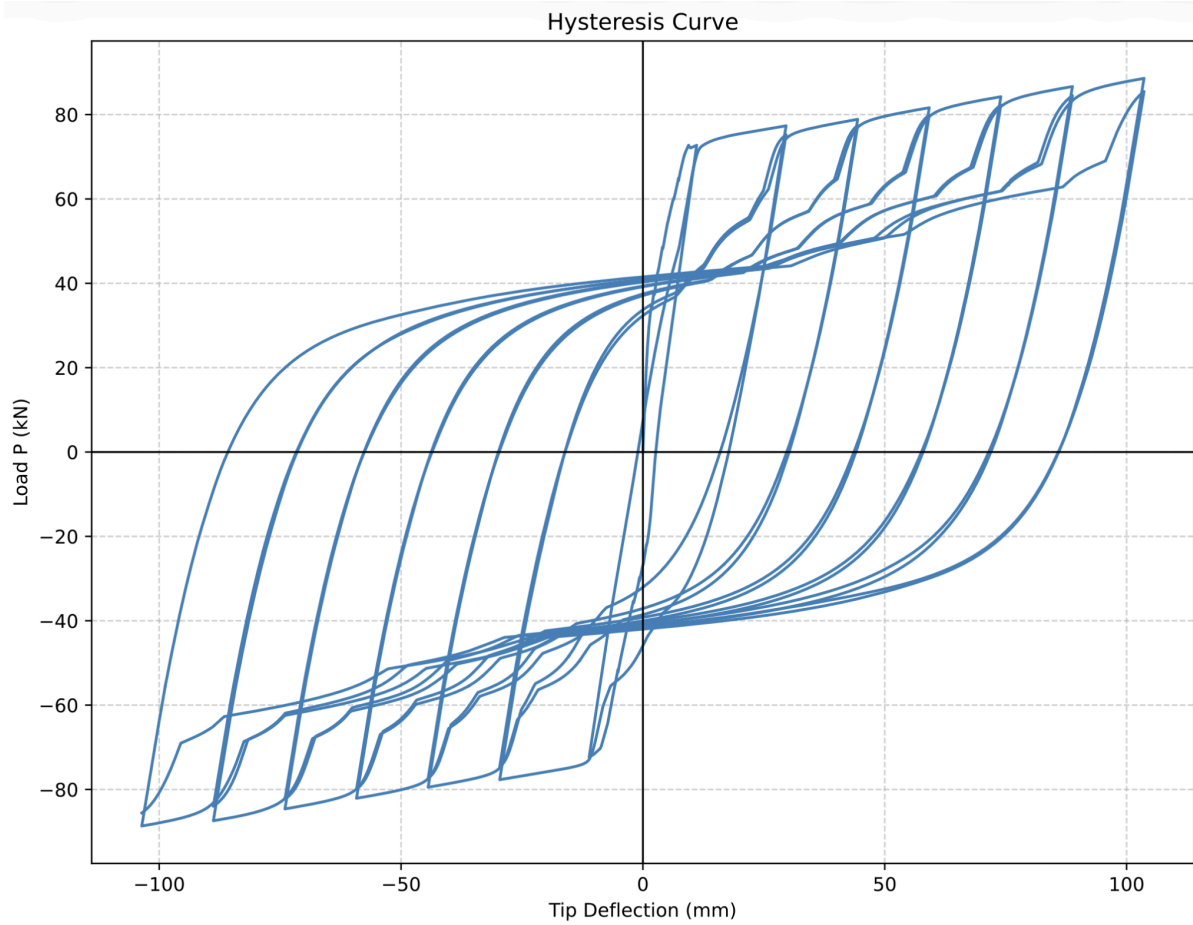


Figure 4. Numerical Response of Model 0

Effects of Element Size

Assignment: In Model 0, decrease the element length to half and then increase to double the initial length – this is especially for the plastic hinge elements. Compare the analysis results with the baseline model and the experimental results and state your observations.

Codes:

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/Model_0_Halved.py

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/Model_0_Halved_plotting.py

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/Model_0_Doubled.py

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/Model_0_Doubled_plotting.py

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/Comparison_plot.py

Observations & Conclusions

According to Paulay and Priestley formula,

$$L_{ph \text{ beam}} \approx 0.08 (2119 - 203) + 0.022 * 295 * 16 \text{ (excluding the rigid link length)} = 257.12 \text{ mm}$$

$$L_{ph \text{ column}} \approx 0.08 (1088.5 - 228.5) + 0.022 * 500 * 16 \text{ (excluding the rigid link length)} = 228.8 \text{ mm}$$

This yields a physical plastic hinge length of 257.12 mm for the beam and 228.8 mm for the column.

$$L_{ph \text{ beam, Model 0}} = 228 \text{ mm}$$

$$L_{ph \text{ column, Model 0}} = 203 \text{ mm}$$

$$L_{ph \text{ beam, Model 0 Doubled}} = 456 \text{ mm}$$

$$L_{ph \text{ column, Model 0 Doubled}} = 406 \text{ mm}$$

$$L_{ph \text{ beam, Model 0 Halved}} = 114 \text{ mm}$$

$$L_{ph \text{ column, Model 0 Halved}} = 101.5 \text{ mm}$$

Double Plastic Hinge Length (Figures 6 & 7): Doubling the plastic hinge element length results in a force-deformation response that remains highly stable and closely tracks the baseline model (Model 0). The peak lateral strength and overall envelope are preserved, though spreading the nonlinearity over an artificially larger element slightly smooths out the local curvature distribution.

The baseline Model 0 element lengths (228.5 mm for the beam, 203 mm for the column) are close to these analytical physical lengths. Thus, Model 0 is numerically stable and a realistic envelope is traced. Doubling the length over-extends this zone. While the numerical stability is maintained, it spreads the local damage over this zone.

Half Plastic Hinge Length (Figure 5): Reducing the plastic hinge element length to half leads to severe numerical instability. The analysis fails to converge and produces an unrealistic load spike of more than 5×10^6

kN at a deflection of approximately -100 mm, completely corrupting the hysteresis response.

Halving the element length forces it to be significantly shorter than the actual physical plastic hinge zone. In a displacement-based formulation, this leads to an extreme strain localization issue. As dispBeamColumn elements use fixed shape functions, the structural response does not converge upon refining the mesh; instead, damage simply concentrates in the smallest available element. Thus, to satisfy the imposed global displacement, the model forces an unrealistically large curvature into this short element. This results in excessive material strains, numerical instability, and ultimately causes the analysis to diverge, producing the unphysical response shown in Figure 7.

Model 0 With Half Plastic Hinge Length

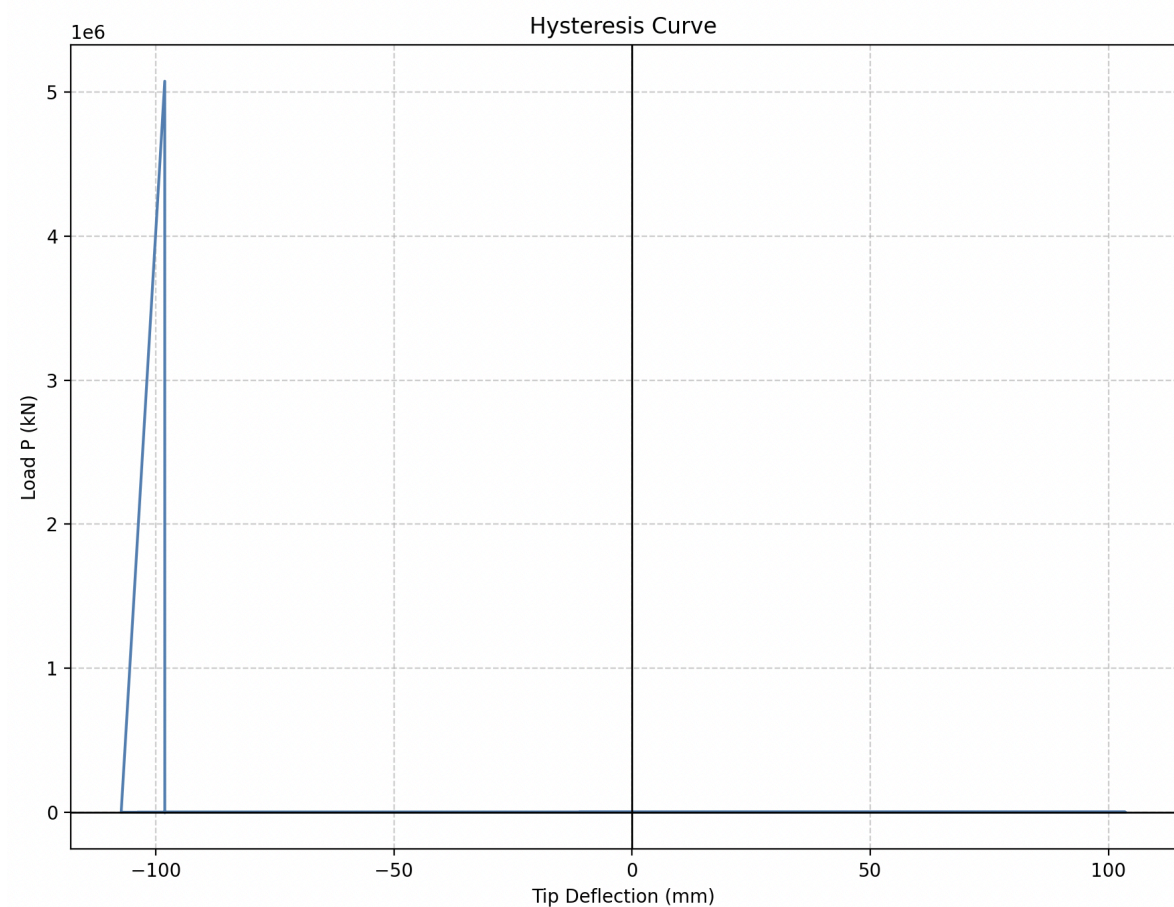


Figure 5. Numerical Response of Model 0 With Half Plastic Hinge Length

Model 0 With Double Plastic Hinge Length

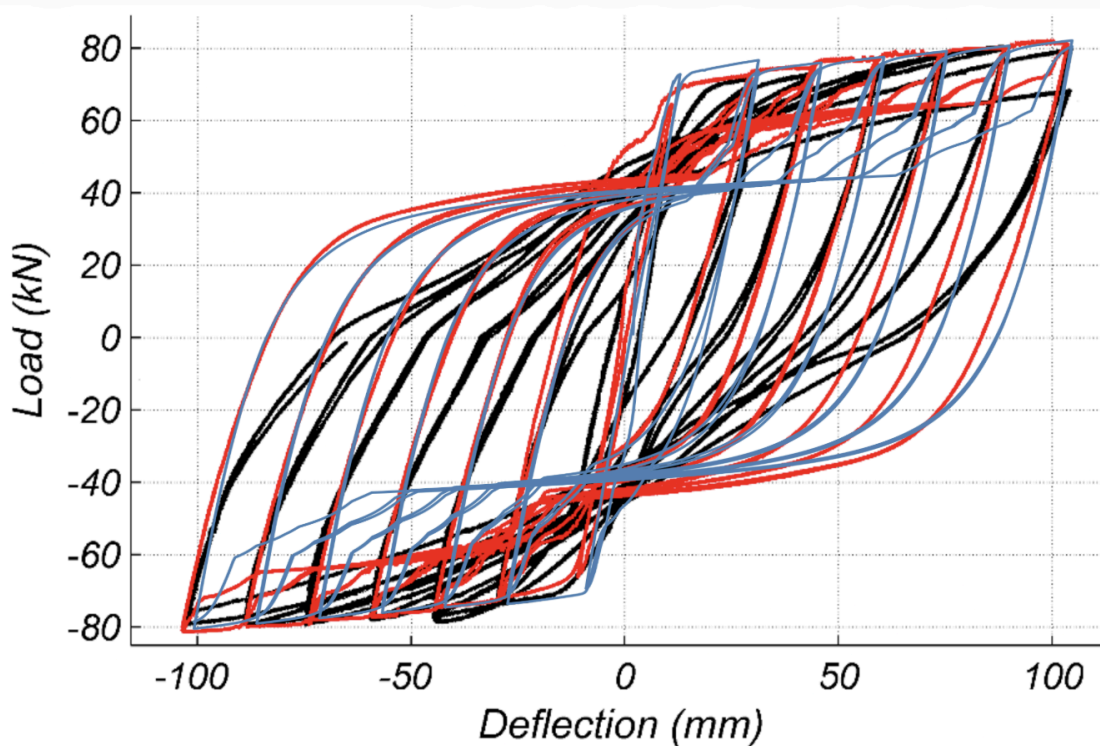


Figure 6. Expected numerical (red), Observed numerical (blue), Experimental (black) correlation of Model 0 With Double Plastic Hinge Length

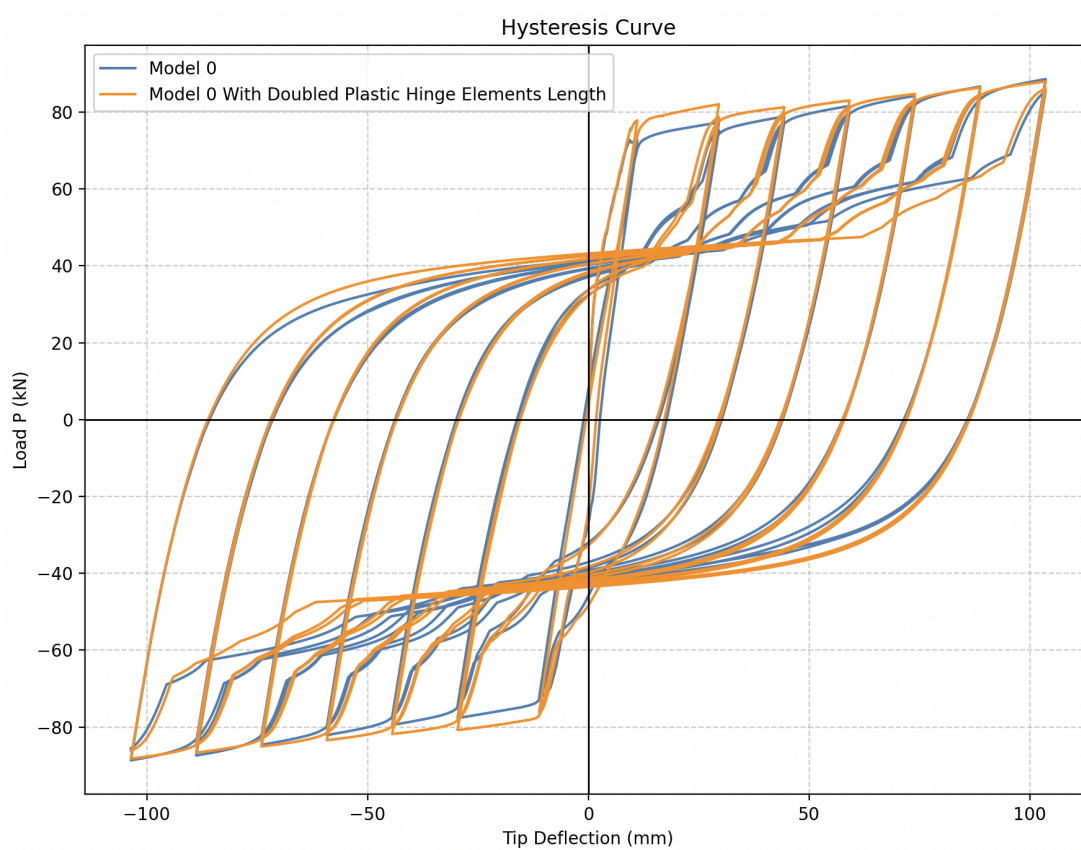


Figure 7. Numerical Response of Model 0 vs Model 0 With Double Plastic Hinge Length

Assignment: In Model 0, decrease the element length to half and then increase to double the initial length – this is especially for the plastic hinge elements. Compare the analysis results with the baseline model and the experimental results and state your observations. Try to investigate the spread of nonlinearity in the first two consecutive beam elements (e.g., the first two beam elements on the left side). For this, you can plot the section moment-curvature or the stress-strain response of a rebar at the integration points of each element. What can you conclude?

Codes:

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/Model_0_Halved.py

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/Model_0_Halved_plotting.py

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/Model_0_Doubled.py

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/Model_0_Doubled_plotting.py

Spread of nonlinearity in the first two consecutive left beam elements

Beam Nodes: 33, 32, 31, 30, 0, 20, 21, 22,

Elements are defined from left to right to maintain geometric transformation

Element 3031 : plastic-hinge element directly adjacent to the left of column joint (nodes 31 to 30)

Element 3132 : next element further from the joint in the left beam (nodes 32 to 31)

The Moment-Curvature Response of First (Section 1) and Last (Section 5) sections are quite similar.

Observations

Model 0

- **Plastic hinge element adjacent to the beam-column joint:** The integration point located at the beam-column joint interface exhibits severe nonlinear behavior. Wide and fully developed hysteretic loops are observed in both the section moment-curvature and reinforcement stress-strain responses, indicating significant yielding and plastic deformation.
- **Adjacent beam element:** Moving away from the joint, the hysteretic loops become much narrower, indicating a rapid reduction in plastic demand. Almost no plasticity is transferred across the boundary between the two elements, as shown in Figure 8(a).

Model 0 Doubled (Double Plastic Hinge Length)

- **Plastic hinge element adjacent to the beam-column joint:** This element is twice as long as in Model 0. Significant plastic deformation still occurs at the beam-column joint interface. However, because the element is longer, the integration points are spaced farther apart. As a result, the model predicts yielding at locations farther from the joint compared with Model 0.
- **Adjacent beam element:** The neighboring beam element again exhibits a rapid reduction in plastic demand. The hysteretic loops are much narrower, while the reinforcement remains essentially

elastic, as indicated by the nearly linear stress-strain response (Figure 9(b)). The boundary between the two elements continues to prevent the spread of plasticity, as shown in Figure 9(a).

Conclusions

The results show that the plastic deformation remains confined within the first yielding element. In Model 0, the damage is concentrated within a 228 mm plastic hinge element, whereas in the doubled model it is confined within a 457 mm element. Although the length of the localized damage zone changes, the localization behavior remains the same.

This behavior is a consequence of strain localization in displacement-based beam-column elements. Once the integration point at the beam-column interface yields, its stiffness decreases significantly. To satisfy the imposed global displacement, the numerical model concentrates most of the additional curvature and rotation within this softened region rather than allowing plasticity to spread into the adjacent element.

Consequently, the plastic hinge remains confined to the interface element, with very limited transfer of inelastic deformation across element boundaries. These results demonstrate that displacement-based formulations are sensitive to mesh discretization and do not naturally reproduce the gradual spread of cracking and yielding unless the plastic hinge length is carefully calculated and represented in the finite element mesh.

Responses of Model 0

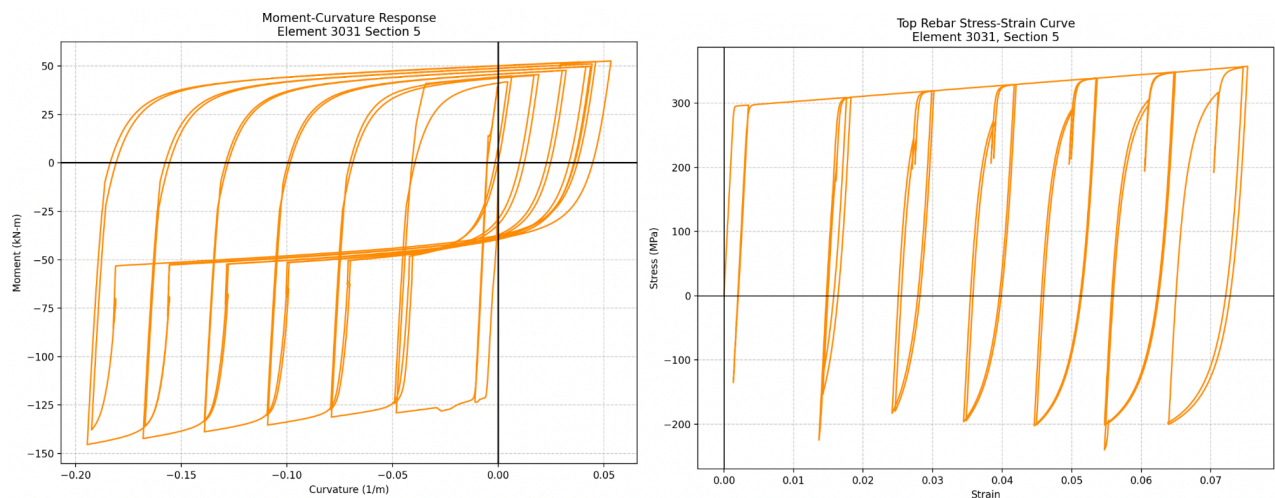


Figure 8(a).

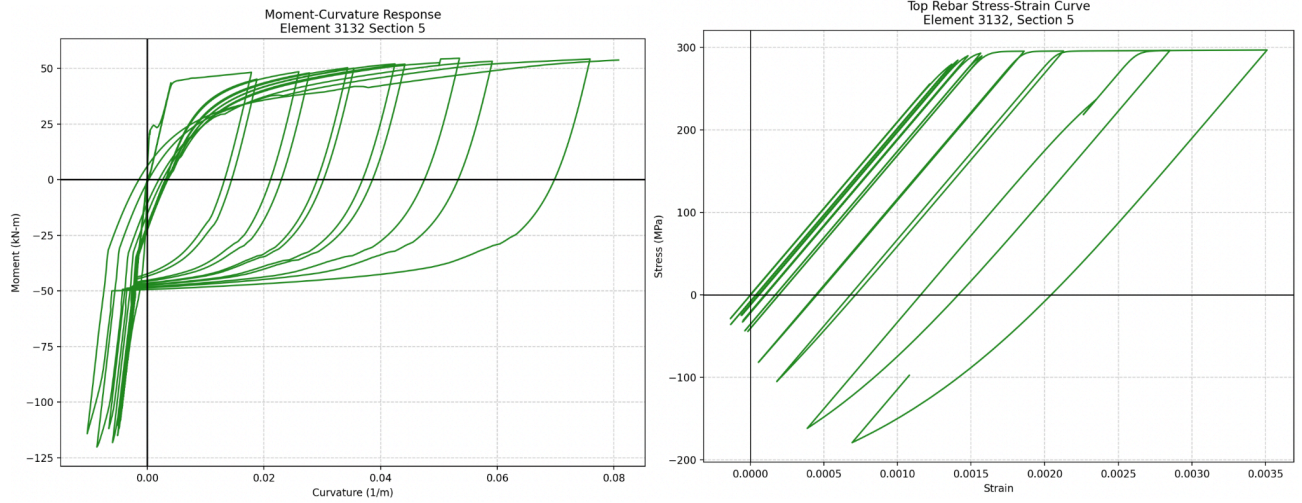


Figure 8(b).

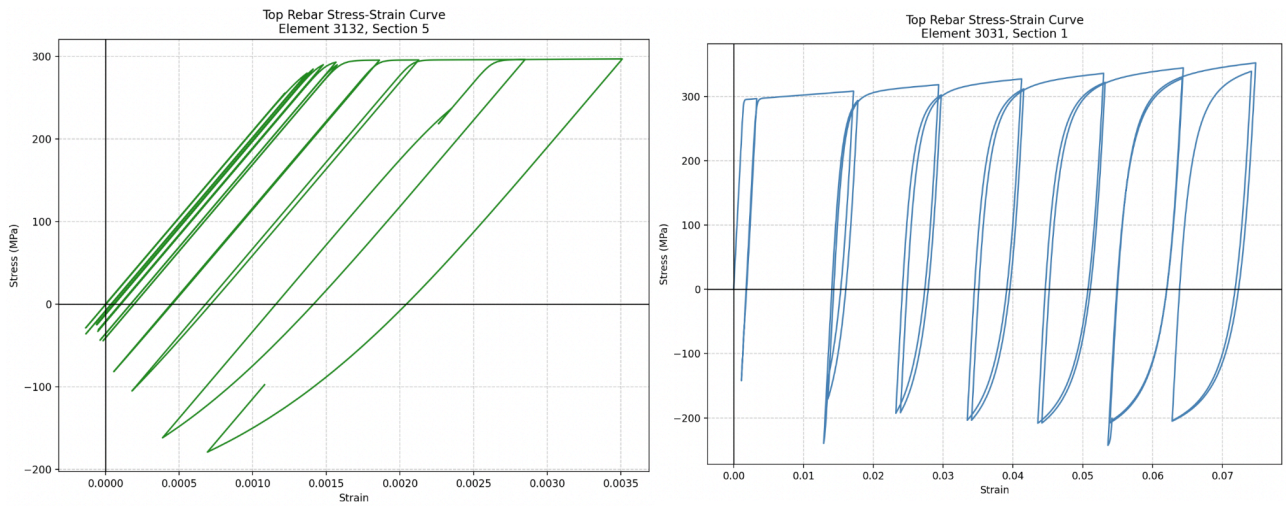


Figure 8(c).

Figure 8. Section Moment-Curvature and Stress-Strain response of corner rebar of Model 0

Responses of Model 0 With Double Plastic Hinge Length

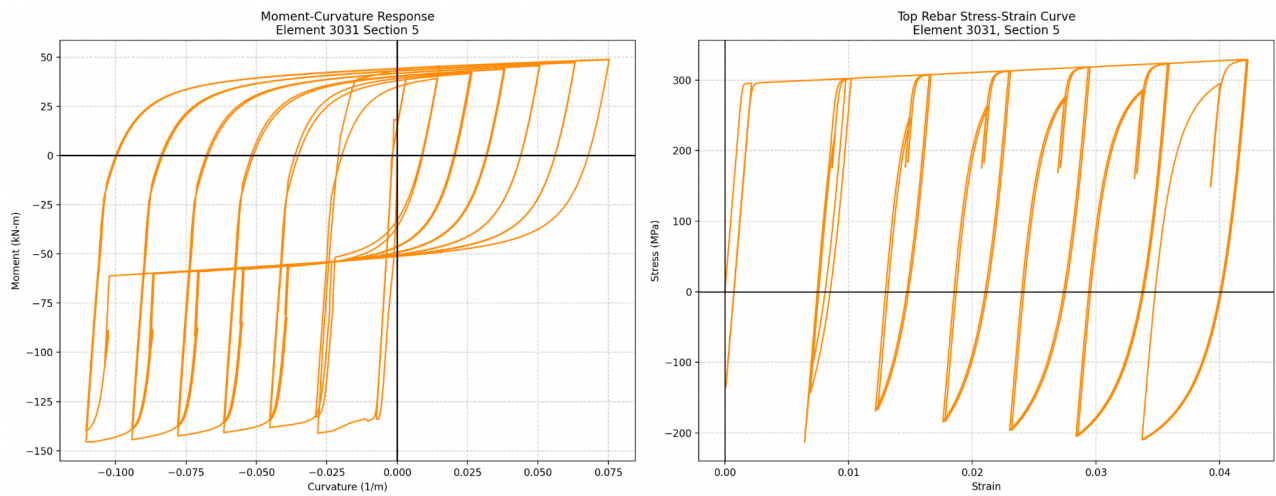


Figure 9(a).

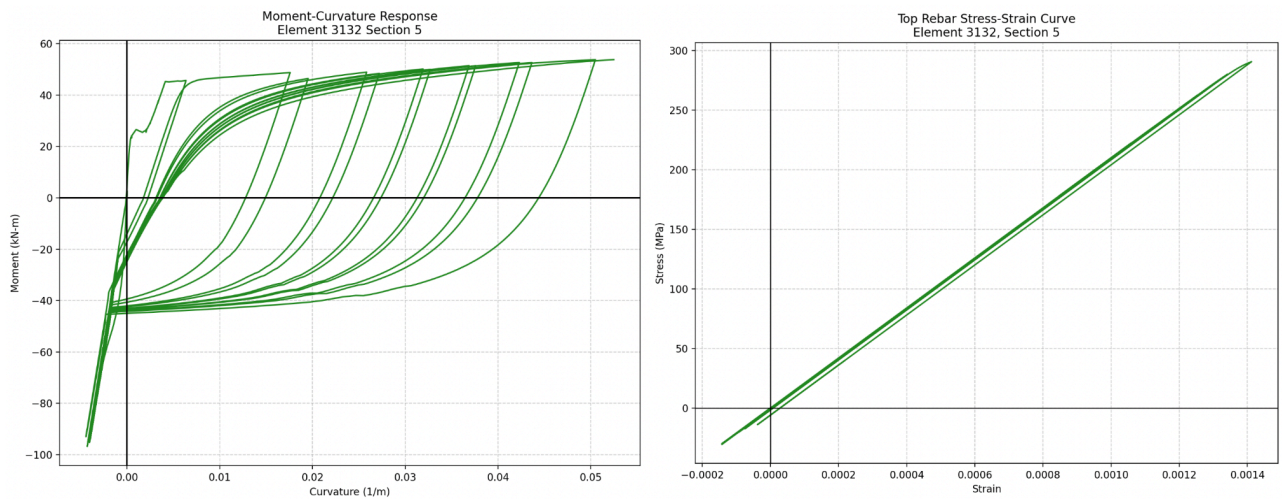


Figure 9(b).

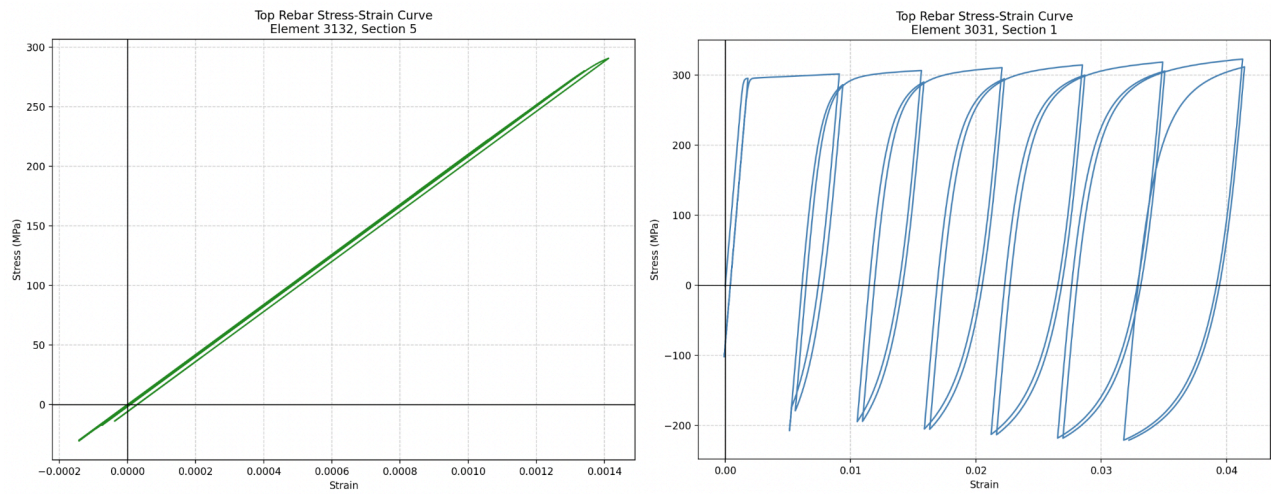


Figure 9(c).

Figure 9. Section Moment-Curvature and Stress-Strain response of corner rebar of With Double Plastic Hinge Length

Force-based vs. displacement-based elements

Assignment: Reconfigure Model 0 with force-based elements. Note that you likely need to use at least two elements for each beam due to the application of gravity load. Adjust the number of integration points and the integration rule so that the tributary length of the first integration point (i.e., the plastic hinge) is approximately one half of the depth of the associated section. Investigate the spread of nonlinearity in the consecutive integration points in the beam. What can you conclude?

Codes:

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/Model_0_force_based.py

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/Model_0_force_based_plotting.py

Calculation of No of Integration Points,

Element Section	Centerline Length	Rigid Link Length	Element Length excluding rigid link	Target Plastic hinge length (L_{ph})	Selected IPs (n)	Tributary Length of IP
Inner Beam	848 mm	203 mm	645 mm	228 mm	4	178.3 mm
Outer Beam	1271 mm	0 mm	1271 mm	228 mm	5	219.5 mm
Column	1236 mm	228 mm	1008 mm	203 mm	5	174.1 mm

The Gauss-Legendre integration rule places all integration points strictly inside the element, with no points located at the element boundaries. In frame structures subjected to lateral loading, the maximum bending moments occur at the beam-column joint faces, which coincide with the element boundaries. Since the integration points are located slightly away from these critical regions, the peak bending moment is not captured accurately.

In contrast, the Gauss-Lobatto integration rule places integration points directly at the element boundaries. This allows the numerical model to capture the peak bending moments at the beam-column joint faces more accurately. Thus, only Gauss-Lobatto is used.

Observations & Conclusions:

Comparing the Model 0 numerical response obtained using the forceBeamColumn element with the experimental hysteresis curve shows good agreement with the expected behavior (Figure 10). Similar to the dispBeamColumn model, the force-based formulation accurately predicts the overall lateral response and peak lateral strength of the beam-column subassembly. However, discrepancies remain in the shape and width of the hysteresis loops, primarily due to the constitutive behavior of the Steel02 and Concrete02 material models.

Compared with the displacement-based formulation, the forceBeamColumn element exhibits significantly fewer convergence problems during the nonlinear cyclic analysis.

Unlike the dispBeamColumn formulation, where the plastic hinge length must be represented by physically sizing the element adjacent to the joint, the forceBeamColumn formulation allows an entire member to be modeled using only a few elements. The plastic hinge length is controlled numerically through the locations and weights of the integration points rather than the finite element mesh.

The section responses demonstrate a smooth and continuous spread of nonlinearity along the member. The integration point located at the beam-column interface (Section 4) experiences the highest plastic demand, exhibiting wide hysteretic loops and significant energy dissipation. Moving away from the joint toward Sections 3, 2, and 1, the hysteretic loops progressively narrow. Consequently, the reinforcement stress-strain responses transition gradually from severe yielding to mild yielding and finally to nearly elastic behavior (Figure 12).

Using only two force-based elements per beam, the model reasonably reproduces the experimental response while allowing yielding to spread naturally across the integration points within the high-moment region near the joint. This behavior is achieved without the severe convergence problems observed in the displacement-based formulation.

The force-based formulation largely eliminates the mesh sensitivity and strain localization associated with displacement-based elements. Rather than assuming a displacement field, force-based elements satisfy internal force equilibrium throughout the member. When the critical section at the beam-column interface yields and loses stiffness, the adjacent integration points continue to resist the required bending moments. If these moments exceed the section yield capacity, yielding develops naturally at the neighboring integration points, resulting in a realistic spread of plasticity.

Furthermore, by selecting an integration rule whose first integration point represents a tributary length approximately equal to half the section depth, the physical plastic hinge length can be directly incorporated into the numerical integration scheme. This provides a physically realistic, and numerically stable simulation of both local damage evolution and the global cyclic response.

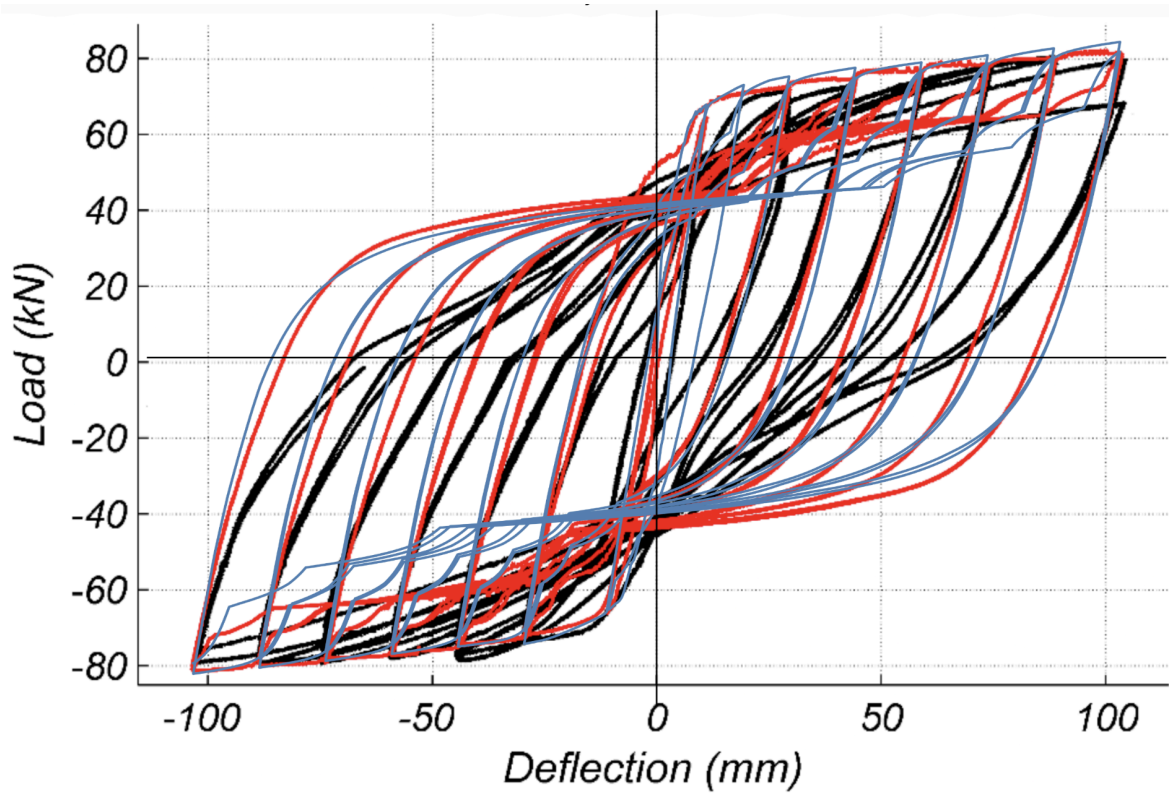


Figure 10. Expected numerical (red), Observed numerical (blue), Experimental (black) correlation of Model 0 With Force-based elements

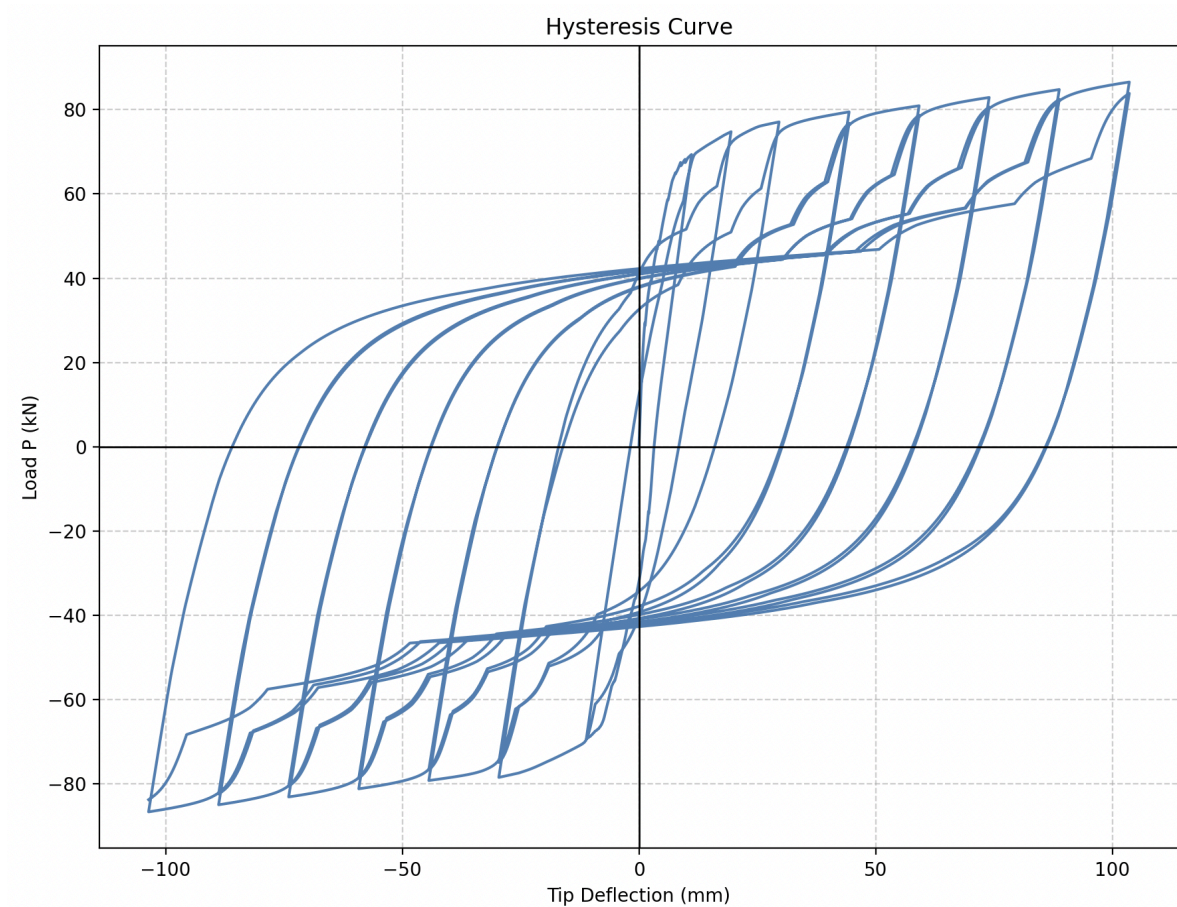


Figure 11. Numerical Response of Model 0 With Force-based elements

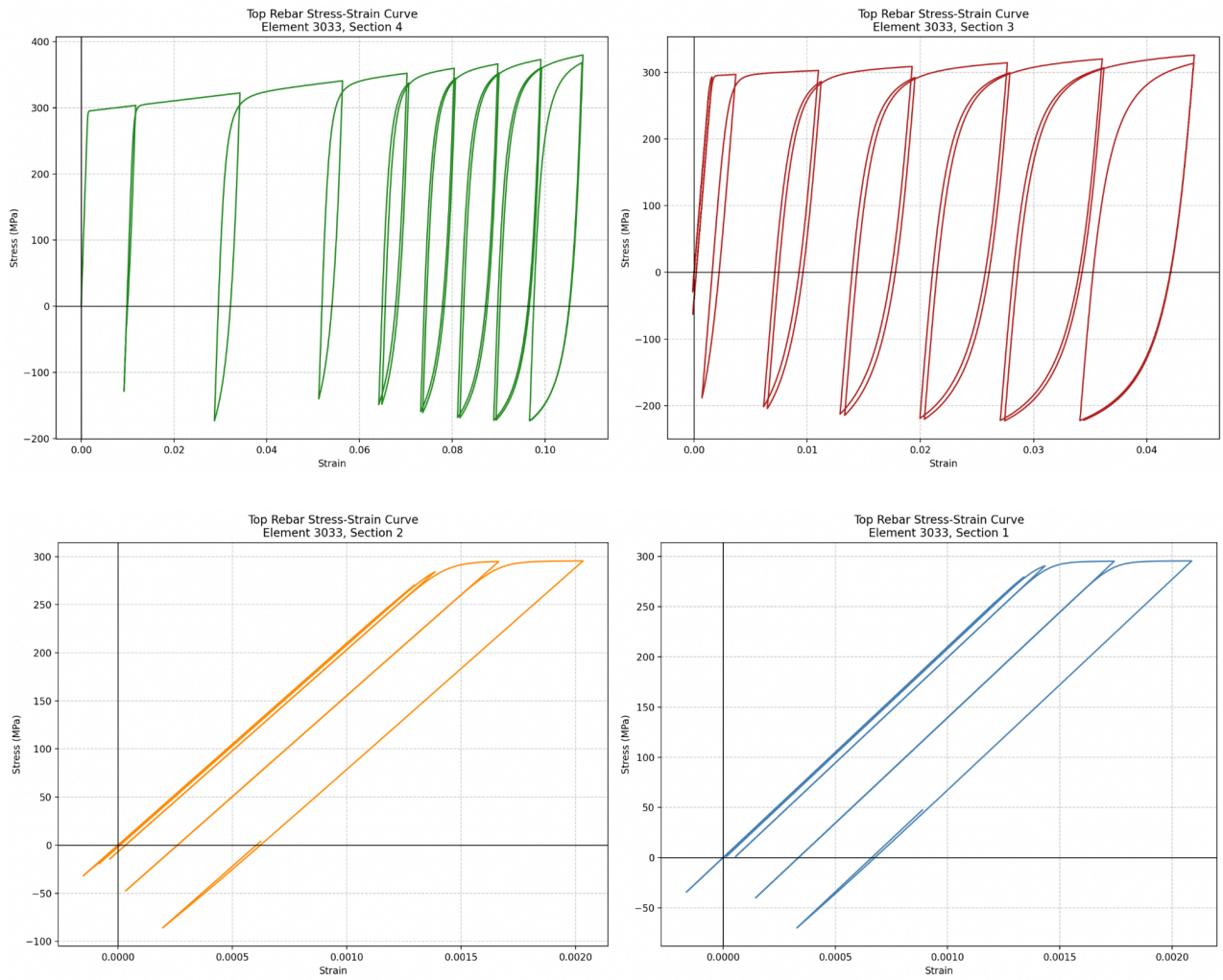


Figure 12. Stress-Strain response of corner rebar of Model 0 With Force-based elements

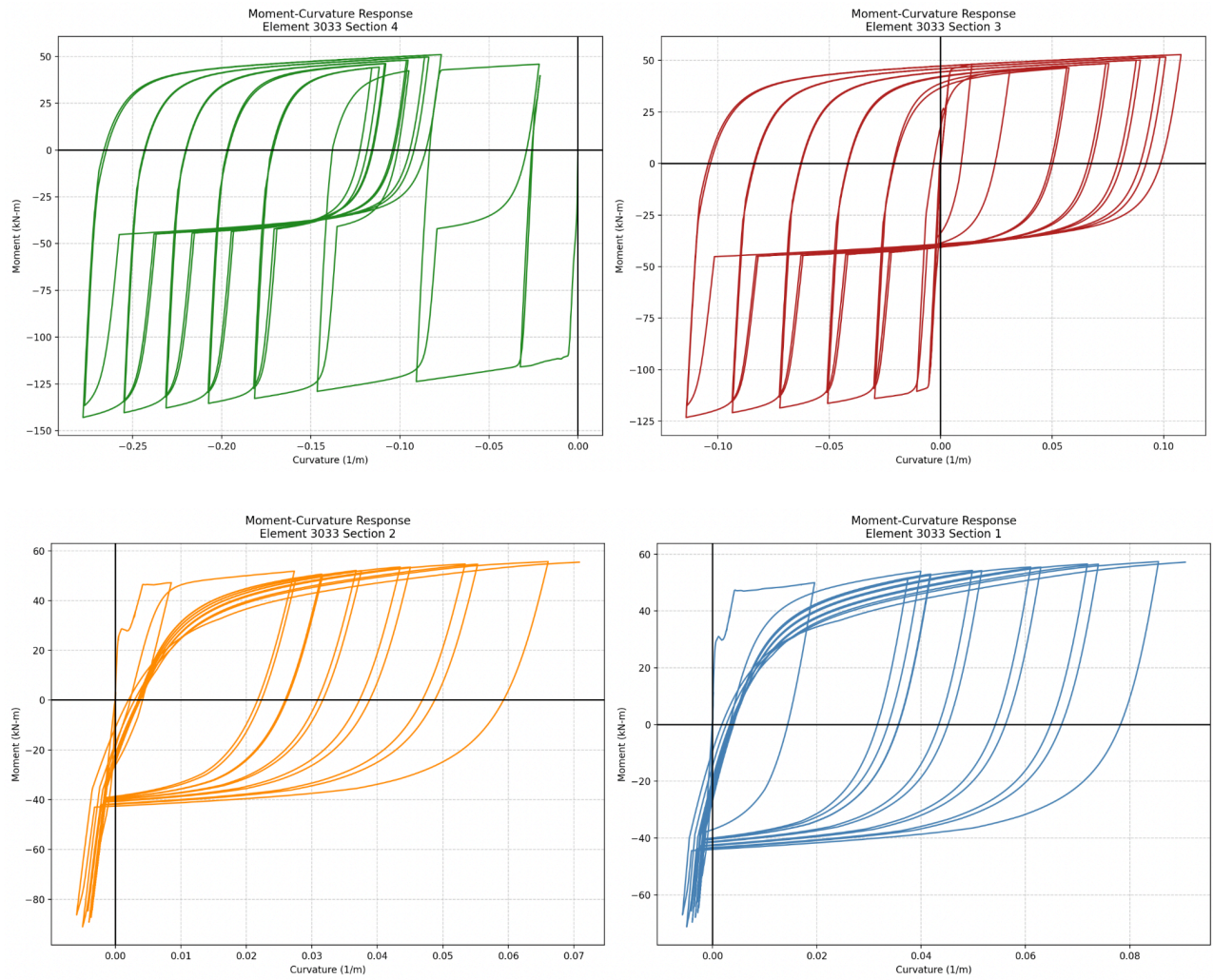


Figure 13. Section Moment-Curvature response of Model 0 With Force-based elements